Syllabus

Statistical Learning

Instructor

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Course Description

The wealth of observational and experimental data available provides great opportunities for us to learn more about our world. This course teaches modern statistical methods for learning from data, such as, regression, classification, kernel methods, and support vector machines.

Course Outcomes

Upon successful completion of this course, you will be able to:

- List modern statistical techniques for modeling and drawing inferences from large data sets.
- Code visual and numerical diagnostics to assess the soundness of their models.
- Explain the computational requirements and compromises to be made in analyzing large data sets.
- Analyze real data sets and communicate their results.
- Implement and use these numerical methods in Python.
- Identify and utilize concepts from computational mathematics and statistics for real-world problem-solving.

Course Materials

The link to reading materials and resources to learn on the topics can be found in each week's learning module. All materials are available online for free, no required resources need to be purchased. There is no required textbook to supplement the course materials. Note: Be aware that some resources may open in a new tab.

Required Textbook: T. Hastie, R. Tibshirani and J. Friedman, *The Elements of Statistical Learning*, 2nd Ed. Springer (2009), ISBN 978-0-38784857-0.

Software Requirements: Python 3.12 or higher along with compatible versions of numpy, pandas, matplotlib, scipy, and scikit-learn

Course Outline

The course consists of 8 modules that focus on the following key areas:

Module 1: Statistical Learning: Terminology and Ideas

Key concepts

- What is Statistical Learning?
- Terminology and types of data
- Formal description of statistical learning.

Readings

- Pg 9-11, 18-22 of the textbook
- Statistical Learning Theory

Module 2: Linear Regression Methods

Key concepts

- Linear Regression and Least Squares
- Subset Selection
- Coefficient Shrinkage for Linear Regression: Ridge Regression and LASSO'
- Data Transformation

Readings

- Pg 11-14; 44-73 of the textbook
- Provided Jupyter Notebook on Coding

Module 3: Linear Classification Methods

Key concepts

- Linear Regression of an Indicator Matrix
- Linear Discriminant Analysis (LDA)
- Logistic Regression

Readings

- Pg 101-129 of the textbook
- Provided Jupyter Notebook on Coding

Module 4: Basis Expansion Methods

Key concepts

- Piecewise Polynomials
- Smoothing Splines
- Regularization via Reproducing Kernal Hilbert Spaces

Readings

- Pg 139-174 of the textbook
- Provided Jupyter Notebook on Coding

Module 5: Kernel Smoothing Methods

Key concepts

- Kernel Smoothers
- k-Nearest Neighbors
- Local Regression

Readings

- Pg 191-205 of the textbook
- Provided Jupyter Notebook on Coding

Module 6: Model Assessment and Selection

Key concepts

- Bias, Variance and Model Complexity
- Bayesian Approach and BIC
- Vapnik-Chernovenkis (VC) Dimension
- Cross Validation
- Bootstrapping

Readings

- Pg 219-254 of the textbook

Module 7: Maximum Likelihood Inference

Key concepts

- Maximum Likelihood Inference
- Bayesian Inference

Readings

- Pg 261-272 of the textbook
- Provided Jupyter Notebook on Coding

Module 8: Advanced Topics

Key concepts

- Additive Models and Trees
- Support Vector Machines
- K-Means Clustering
- Neural Networks

Readings

- Pg. 295-317; 389-409; 417-438; 459-475 of textbook
- Provided Jupyter Notebook on Coding

Course Structure and Learning Activities

There are 8 content modules in this course and each module may take about 9-12 hours to complete. A consistent pace will help you complete the module and move on to the next course in the sequence. The final module consists of your final exam for the course.

This course is comprised of the following elements:

- **Readings**: Each module may include several required and/or supplemental readings.
- **Video Lessons**: In each module, the concepts you need to know will be presented through a collection of short videos. You may stream these videos for playback within the browser by clicking on their titles.
- **Discussion Forum**: This course has a place for you to interact with other learners about the content in Module 7, where we tackle maximum likelihood inference and Bayesian inference. This is a graded activity in the class.
- **Practice Quizzes**: Each module will include some practice quizzes, intended for you to assess your understanding of the topics. You will be allowed unlimited attempts at each practice quiz. There is no time limit on how long you take to complete each attempt at the quiz. These quizzes do not contribute toward your final score in the class.
- **Summative Module Assessments:** Each module will include at least one summative module assessment. You will be allowed one attempt every eight hours for each assessment. There is no time limit on how long you take to complete each attempt at the assessment. Your highest grade will be recorded.
- **Final Assessment:** This course will contain one summative course assessment. You will be allowed one attempt for the assessment. Before taking the exam, please make sure you are in a place with reliable internet connection. No retakes will be granted for the lack of internet access. You are in an online program and the use of the Internet is a requirement.

How to Pass This Course

Guidelines for completing and submitting each assigned course activity is posted along with the assignment. Assignments can be submitted at any time as you move through the module. Only those who complete and submit all assignments, including peer reviews, will receive a certificate of completion of this course. *No late assignments will be accepted.* In case of extenuating circumstances beyond your control that prevent the submission of an assignment or exam, you have to enter a request with the program advisor and the instructor.

To qualify for a Course Certificate, simply start verifying your coursework at the beginning of the course and pay the fee. Coursera Financial Aid is available to offset the registration cost for learners with demonstrated economic needs. If you have questions about Course Certificates, please see the help topics here.

Also, note that this course is part of Master of Data Science offered by Illinois Institute of Technology. By earning a Course Certificate in this course, you are on your way toward earning a Specialization Certificate in this topic. See more information about this program here.

If you choose not to pay the fee, you can still audit the course. You will still be able to view all videos, submit practice quizzes, and view required assessments. Auditing does not include the option to submit required assessments. As such, you will not be able to earn a grade or a Course Certificate.

The following table explains the breakdown for what is required in order to pass the class and qualify for a Course Certificate. You must pass each and every required activity in order to pass this course.

Activity	Required?	Number per Course	Estimated Time per Module	% Required to Pass	% of Total Grade
Lecture Videos	Yes	3-6 per module	.5-1 hour	N/A	N/A
Practice Quizzes	No	3-6 per module	.5 hour	N/A	N/A
Discussions	Yes	1-2 per course	1 hour	N/A	N/A
Summative Module Assessments	Yes	1 per module	.5 hour	80%	7.5%/each module (60%)
Summative Course Assessment	Yes	1-2 per course	1-3 hours	80%	40%

Letter Grades

Letter grades are used for the final grade. Information about IIT grading system can be found in the Graduate Student Handbook.

Letter Grade	Description	Points	Percentage
A	Excellent	4.00	90-100
B	Above Average	3.00	80-89.99
C	Average	2.00	70-79.99
E	Fail	0.00	Under 70%

Getting and Giving Help

- Use the Learner Help Center to find information regarding specific technical problems. For example, technical problems would include error messages, difficulty submitting assignments, or problems with video playback. If you cannot find an answer in the documentation, you can also report your problem to the Coursera staff by clicking on the *Contact Us!* link available on each topic's page within the Learner Help Center.
- Use the flag icon under each item to report errors in lecture video content, assignment questions and answers, assignment grading, text and links on course pages, or the content of other course materials.
- Familiarize yourself with Coursera's policy on Accessibility.

Academic Integrity

Your attentiveness to academic integrity reflects the value you place on your own work and the work of others. In addition to Coursera's Honor Code, we also have high expectations for conduct during course participation.

Discussion Forums: Expectations

Sharing an online course with other avid learners like you gives you a unique opportunity to share, collaborate, and learn from others and their experiences, and helps you reinforce your understanding of the topics of the course. Interacting in the Discussion Forums is a great way to engage with your online community. We know that it is not possible to read every discussion forum post, so we recommend that you read those that interest you; and reply when you can contribute. The forum is part of your class activities and everybody is expected to act professionally and be civil and respectful of others in your class. Failure to meet these expectations may be considered a break in the Academic Code of Conduct and may result in your removal from the course. Please, check tips and helpful tools to interact in discussion forums in this document.

Academic Code of Conduct

Above all else, learners are expected to ensure that their conduct helps to create an atmosphere conducive to learning and the interchange of knowledge. While it is understood that some of these items are subject to interpretation, learners should nonetheless endeavor to:

- Be respectful of fellow learners.
- Not discriminate against fellow learners in any manner.
- Conduct peer reviews in a timely manner and give useful feedback on what was done well, helpful suggestions for how to improve, and specific comments about why you gave the grade you chose to assist peers in their learning.